

Sentiment Analysis of Banglish (Bangla–English Code-Mixed) Text Using Bag-of-Words and TF-IDF Representations with Naïve Bayes and Logistic Regression

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Abstract—This paper presents a comparative study of machine learning approaches for sentiment classification of Banglish text (Bengali expressed in Roman/English script and frequently mixed with English words as commonly used on Bangladeshi social media). A dataset of 25,000 Banglish sentences labeled negative, neutral, or positive was preprocessed by normalizing emojis into descriptive English tokens, removing punctuation, filtering a custom list of 94 Banglish stopwords, and discarding documents left empty after cleaning, leaving 24,713 usable samples. Two text representations, Bag-of-Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF), were each combined with two classifiers, Naïve Bayes and multinomial Logistic Regression (trained with cross-validated regularization via *glmnet*), using an 80/20 stratified train–test split. Naïve Bayes performed poorly on both representations, with accuracies of 31.6% (BoW) and 31.5% (TF-IDF), at or below the majority-class baseline of approximately 34.4%. In contrast, Logistic Regression achieved near-perfect performance on both representations, with accuracy, precision, recall and F1-score all above 99.7%. These results indicate that the dataset's sentiment classes are lexically well separated and that a discriminative, regularized linear classifier is far better suited to this task than a Gaussian-assumption Naïve Bayes classifier. We analyze the likely causes of the Naïve Bayes underperformance, the near-identical results of BoW and TF-IDF under Logistic Regression, and outline directions for future work, including transformer-based models and evaluation on noisier, real-world Banglish text.

Keywords—Banglish, code-mixed text, sentiment analysis, Bag-of-Words, TF-IDF, Naïve Bayes, Logistic Regression, natural language processing.

I. INTRODUCTION

Code-mixing the practice of blending two or more languages within the same sentence or conversation is a widespread phenomenon in multilingual societies. In Bangladesh, a large share of informal digital communication takes place in “Banglish,” where Bengali words are transliterated into the Roman alphabet and freely mixed with English words, abbreviations, and emojis. This style dominates comments, reviews, and posts on platforms such as Facebook, YouTube, and e-commerce sites, making it an important but challenging target for sentiment analysis.

Unlike standard English or formal Bengali text, Banglish has no fixed spelling convention: the same word may be transliterated in several different ways, sentence structure often mixes grammar from both languages, and emojis are frequently used to carry sentiment that is not explicit in the words themselves. These properties make standard, off-the-shelf sentiment analysis pipelines built for monolingual text less reliable, and motivate dataset- and language-specific preprocessing.

This paper investigates two classical, computationally inexpensive approaches to representing Banglish text Bag-of-Words (BoW) and TF-IDF and evaluates how well two standard classifiers, Naïve Bayes and Logistic Regression, can classify Banglish sentences into negative, neutral, or positive sentiment. The main contributions of this work are:

- 1) A preprocessing pipeline tailored to Banglish text, including a custom emoji-to-text mapping and a hand-curated list of 94 Banglish stopwords.
- 2) A controlled comparison of Bag-of-Words and TF-IDF representations under identical preprocessing and train–test splits.
- 3) A comparison of a generative classifier (Naïve Bayes) and a discriminative, regularized linear classifier (Logistic Regression) on the same features.
- 4) An analysis of why Naïve Bayes underperforms dramatically relative to Logistic Regression on this dataset, with implications for future Banglish sentiment analysis work.

II. LITERATURE REVIEW

Sentiment analysis of Bangla-English code-mixed text has received growing attention as social media usage in Bangladesh has expanded. Hossain et al. [1] studied sentiment classification of Bangla-English code-mixed and transliterated social media comments, explicitly modeling the role of emojis in expressing sentiment. Their pipeline combined TF-IDF and Count Vectorizer feature extraction with a suite of classical machine learning algorithms, showing that careful handling of emojis and transliteration noise substantially affects classification quality a finding consistent with the emoji-normalization step adopted in this paper.

Raihan et al. [2] introduced SentMix-3L, a code-mixed sentiment analysis test set spanning three languages Bangla, English, and Hindi. Their evaluation compared multilingual transformer models with zero-shot large language model prompting, finding that zero-shot GPT-3.5 prompting outperformed fine-tuned transformer baselines on their three-way code-mixed data. This highlights that code-mixed sentiment analysis remains difficult even for modern large models, especially when more than two languages interact.

Shamael et al. [3] released BanglishRev, a large-scale e-commerce review dataset containing 1.74 million Bangla, English, and Banglish reviews. Using review ratings as weak sentiment labels, they trained a BanglishBERT model and reported strong binary sentiment classification performance, demonstrating that large volumes of naturally occurring code-mixed review data can support high-accuracy sentiment models when paired with transformer-based encoders.

Tareq et al. [4] proposed a data-augmentation approach for Bangla-English code-mixed sentiment analysis, comparing Logistic Regression, Decision Tree, Support Vector Machine, and Extreme Gradient Boosting classifiers with Word2Vec, FastText, and a custom cm-FastText embedding. Their results showed that augmenting limited code-mixed training data with cross-lingual word replacement improved classifier performance across all tested machine learning models, underlining the sensitivity of Banglish sentiment models to training data coverage.

Alam et al. [5] introduced BnSentMix, a diverse Bengali-English code-mixed sentiment dataset of 20,000 samples drawn from Facebook, YouTube, and e-commerce sources and annotated with four sentiment labels. They benchmarked 14 baseline methods, including classical machine learning, neural networks, and transformer encoders further pre-trained on code-mixed Bengali-English text, achieving 69.8% accuracy and 69.1% F1-score with their best model figures that illustrate how much harder real-world, multi-source Banglish sentiment classification is compared to single-source or synthetically balanced datasets such as the one used in this paper.

Collectively, this body of work shows that (i) emoji and transliteration handling materially affects Banglish sentiment classification, (ii) both classical machine learning and modern transformer-based methods have been applied to this task with varying success, and (iii) reported accuracies on diverse, real-world Banglish datasets are typically far lower than the near-perfect scores that can be obtained on smaller or more homogeneous datasets a contrast that is directly relevant to the results reported in Section IV of this paper.

III. METHODOLOGY

A. Dataset Description

The dataset used in this study consists of 25,000 Banglish text samples, each labeled with one of three sentiment classes: negative, neutral, or positive. Table I summarizes the

class distribution, which is close to balanced across the three classes.

TABLE I. DATASET CLASS DISTRIBUTION

Sentiment Class	No. of Samples	Percentage
Negative	8,434	33.7%
Neutral	8,042	32.2%
Positive	8,524	34.1%
Total	25,000	100%

The distribution of text length was also examined to characterize the dataset, as shown in Fig. 1 and Fig. 2.

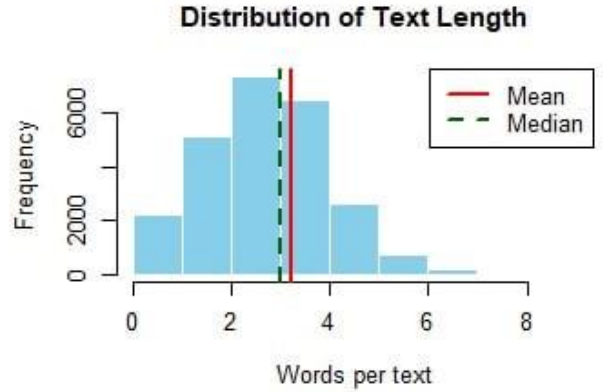


Fig. 1. Distribution of text length.

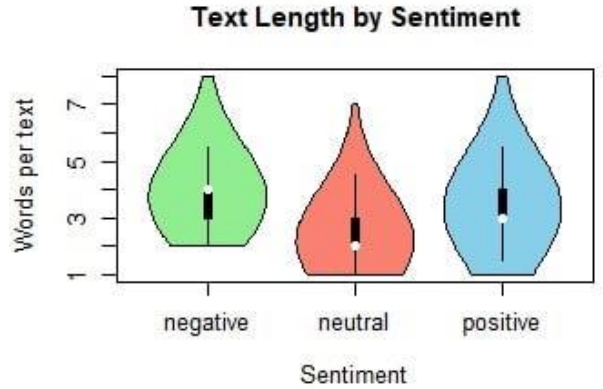


Fig. 2. Text length by sentiment.

B. Data Preprocessing

Raw Banglish text required substantial cleaning before feature extraction. First, a custom dictionary of 20 commonly occurring emojis (e.g., “😄”, “😡”, “❤️”) was used to replace each emoji with a descriptive English token such as “smile”, “angry”, or “love”, preserving the sentiment signal the emoji conveys instead of discarding it. Text was then lowercased and stripped of punctuation. A hand-curated list of 94 Banglish stopwords covering pronouns (“amra”, “tumi”), discourse markers (“kintu”, “tobe”), auxiliary/linking words (“ache”, “hoilo”), and a small set of

generic nouns that appeared with high frequency across all three classes (e.g., “traffic”, “phone”, “class”) was removed using the `tm` package’s `removeWords` transformer, followed by whitespace normalization.

After cleaning, documents that became empty (i.e., contained no remaining terms in the Document-Term Matrix) were discarded. This reduced the dataset from 25,000 to 24,713 samples, a loss of 287 samples (1.15%), and the class distribution was re-verified to remain approximately balanced after this filtering step.

C. Feature Extraction

Two representations were built from the cleaned corpus using the `tm` package’s `DocumentTermMatrix` function. The Bag-of-Words (BoW) representation counts raw term frequencies for each document, while the TF-IDF representation applies term frequency–inverse document frequency weighting, which down-weights terms that occur in a large proportion of documents and up-weights terms that are more discriminative of individual documents. In both cases, `removeSparseTerms` was applied with a sparsity threshold of 0.999, retaining only terms that appeared in at least 0.1% of documents, in order to keep the resulting matrices computationally tractable.

D. Train–Test Split

For both representations, the data was split into 80% training and 20% testing subsets using `caTools`’ `sample.split` function with a fixed random seed (123) for reproducibility. The split was performed independently on the label vector, preserving approximately the original class proportions in both the training and test sets.

E. Classification Models

Two classifiers were trained and evaluated on each representation:

Naïve Bayes: implemented via the `e1071` package’s `naiveBayes` function with Laplace smoothing ($\alpha = 1$). This is a generative classifier that assumes conditional independence between features given the class label. Notably, `e1071`’s implementation models numeric predictors including both raw BoW counts and TF-IDF weights under a Gaussian likelihood assumption, which does not naturally match the sparse, non-negative, highly skewed distributions typical of bag-of-words style features.

Logistic Regression: implemented via the `glmnet` package’s `cv.glmnet` function with `family = “multinomial”`, which fits a regularized multinomial Logistic Regression and uses k-fold cross-validation to select the regularization strength λ that minimizes cross-validated error. Predictions were generated using the `lambda.min` value. This is a discriminative linear classifier that directly models the decision boundary between classes rather than the distribution of features within each class.

F. Evaluation Metrics

Model performance was evaluated on the held-out 20% test set using four metrics computed from the confusion matrix produced by the `caret` package’s `confusionMatrix` function: overall Accuracy, and macro-averaged Precision (mean of per-class Positive Predictive Value), Recall (mean of per-class Sensitivity), and F1-Score (the harmonic mean of the per-class precision and recall, averaged across classes). Macro-averaging was chosen so that performance on each of the three sentiment classes contributes equally to the reported metrics, regardless of class size.

IV. RESULTS AND ANALYSIS

Table II reports the accuracy, precision, recall, and F1-score obtained for each combination of text representation (BoW or TF-IDF) and classifier (Naïve Bayes or Logistic Regression) on the 20% held-out test set. Fig. 3 visualizes the same results alongside the majority-class baseline of approximately 34.4% (the proportion of the most frequent sentiment class, “positive”, in the cleaned dataset).

TABLE II. MODEL PERFORMANCE COMPARISON

Representation	Model	Accuracy	Precision	Recall	F1-Score
BoW	Naive Bayes	0.3161	0.6576	0.3345	0.2432
BoW	Logistic Regression	0.9980	0.9979	0.9980	0.9980
TF-IDF	Naive Bayes	0.3146	0.1576	0.3331	0.4793
TF-IDF	Logistic Regression	0.9980	0.9979	0.9980	0.9980

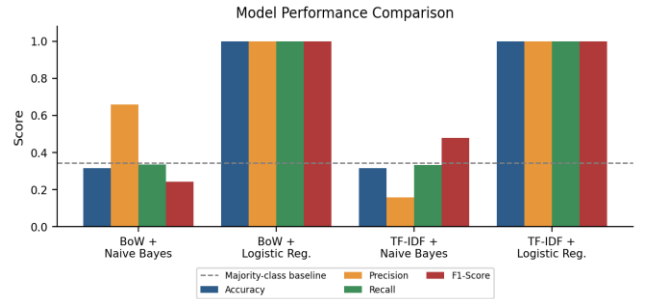


Fig. 3. Accuracy, precision, recall, and F1-score for all four representation–model combinations, relative to the majority-class baseline (dashed line).

The results show a clear and consistent pattern: the choice of classifier, not the choice of text representation, is the dominant factor in performance. Naïve Bayes achieved only 31.6% accuracy with BoW and 31.5% with TF-IDF both at or slightly below the 34.4% majority-class baseline meaning Naïve Bayes performed no better than, and in the BoW case slightly worse than, a trivial classifier that always predicts the most common class. Its precision and recall were also weak and, for TF-IDF, its precision collapsed to 15.8%. By contrast, Logistic Regression achieved 99.80% accuracy on

both BoW and TF-IDF representations, with precision, recall, and F1-score all at or above 99.7%. The confusion matrices confirm this: Logistic Regression correctly classified 4,932 of the 4,942 test samples (BoW), with almost all errors occurring between the neutral and positive/negative classes, while Naïve Bayes' predictions were dominated by a single class, driving its Kappa statistic close to zero (0.0017 for BoW, -4×10^{-4} for TF-IDF) indicating agreement with the true labels barely better than chance.

This large gap has a concrete technical explanation. e1071's naiveBayes implementation treats numeric predictors which includes both BoW term counts and TF-IDF weights as Gaussian-distributed within each class. Bag-of-words style features are, however, sparse, non-negative, and highly skewed (most entries are zero), which is a poor match for a Gaussian assumption; this mismatch is especially severe for TF-IDF, whose continuous, unevenly scaled weights further violate the assumption, which explains why TF-IDF Naïve Bayes' precision (15.8%) was even lower than BoW Naïve Bayes' precision (65.8%). Logistic Regression makes no such distributional assumption about the features; instead, it directly learns a linear decision boundary that separates the classes, and the regularization used via `cv.glmnet` handles the high dimensionality and sparsity of the feature matrices effectively, which is consistent with its near-identical, near-perfect performance under both BoW and TF-IDF.

The consistently high separability under Logistic Regression also suggests that sentiment in this dataset is strongly signaled by a relatively small set of lexical cues (specific Banglish words and normalized emoji tokens), rather than by subtler contextual or syntactic patterns. This is a plausible explanation for why performance is so much higher here than the 69.8% accuracy and 69.1% F1-score reported by Alam et al. [5] on the more diverse, multi-source BnSentMix dataset, and the comparatively modest scores reported in other related work [1], [2], [4] on noisier, real-world code-mixed text. Comparing average accuracy across representations (mean of the Naïve Bayes and Logistic Regression results), BoW attained a marginally higher average (65.71%) than TF-IDF (65.63%); however, this difference is driven entirely by the noisy Naïve Bayes results and is not meaningful given that the two representations perform identically under Logistic Regression, the clearly stronger classifier for this task. Taking accuracy as the selection criterion, BoW combined with Logistic Regression was identified as the best-performing configuration overall.

V. CONCLUSION

This paper compared Bag-of-Words and TF-IDF text representations, each combined with Naïve Bayes and Logistic Regression classifiers, for sentiment classification of Banglish (Bangla-English code-mixed) text. After a preprocessing pipeline that normalized emojis into descriptive tokens and removed a custom list of Banglish stopwords, Logistic Regression achieved near-perfect performance (99.80% accuracy, $\geq 99.7\%$ precision/recall/F1)

under both representations, while Naïve Bayes performed at or below the majority-class baseline ($\approx 31.5\%$ accuracy) under both representations. The results demonstrate that, for this dataset, the choice of classifier matters far more than the choice of feature representation, and that e1071's Gaussian-assumption Naïve Bayes implementation is poorly suited to sparse bag-of-words style features regardless of whether raw counts or TF-IDF weighting is used. Logistic Regression, as a regularized discriminative linear model, is therefore recommended over this Naïve Bayes formulation for Banglish sentiment classification tasks of this kind.

VI. FUTURE WORK

Several directions could extend this work. First, a proper multinomial (event-model) Naïve Bayes implementation rather than e1071's Gaussian-assumption formulation should be evaluated for a fairer comparison against Logistic Regression. Second, the pipeline should be tested on more diverse, real-world, multi-source Banglish data (e.g., BnSentMix [5] or BanglishRev [3]) to check whether the near-perfect Logistic Regression performance generalizes beyond this dataset, or whether it reflects strong lexical separability specific to this data. Third, transformer-based models such as BanglishBERT, as used by Shamalet al. [3], and large language model zero-shot prompting, as explored by Raihan et al. [2], should be benchmarked against the classical methods used here. Fourth, incorporating n-gram features and pretrained word embeddings (Word2Vec, FastText, or cross-lingual embeddings, as explored by Tareq et al. [4]) may improve robustness to Banglish spelling variation. Finally, replacing the single 80/20 train-test split with k-fold cross-validation would provide more statistically robust performance estimates.

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